

Kamrun Nahar Mily

PhD Candidate in Mathematics

Department of Mathematics
University of Houston

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EDUCATIONAL BACKGROUND

Doctor of Philosophy in Applied Mathematics

Spring 2022 – Expected Fall 2026

University of Houston, Houston, Texas, USA

Advisor: Dr. Annalisa Quaini

Research Focus: Agent-Based Modeling for Crowd and Crime Dynamics

Master of Science in Applied Mathematics

2025

University of Houston, Houston, Texas, USA

GPA: 3.770/4.00

Master of Science in Applied Mathematics

2016 – 2017

University of Dhaka, Dhaka, Bangladesh

GPA: 3.97/4.00

A.F. Mujibur Rahman Gold Medal – First Class First

Bachelor of Science in Mathematics

2010 – 2015

University of Dhaka, Dhaka, Bangladesh

CGPA: 3.49/4.00

Project: A Study of Differential Equations in Population Biology

RELEVANT TEACHING EXPERIENCE

University Teaching Positions

- Instructor of Record (MATH 2131: Linear Algebra with MATLAB) University of Houston-
Spring 2026 – Present

- **Graduate Teaching Assistant**

Spring 2022 – Present

Department of Mathematics, University of Houston, Houston, TX

- Led weekly recitation sections and laboratory sessions for Calculus I–II (80+ students/semester)
- Conducted extended office hours and individualized tutoring sessions
- Guided students through proofs, problem-solving techniques, and exam preparation
- Collaborated with faculty on grading, exam design, and development of supplementary resources

- **Lecturer**

August 2017 – July 2021

Department of Mathematics, Primeasia University, Dhaka, Bangladesh

- Primary instructor for 6 undergraduate mathematics courses per semester
- Courses taught: Differential Calculus, Integral Calculus, Business Mathematics, Ordinary Differential Equations, Complex Analysis, Numerical Methods, Linear Algebra
- Designed comprehensive syllabi and delivered interactive lectures for cohorts of up to 80 students
- Developed innovative assessment strategies and coordinated tutorial sessions
- Mentored undergraduate students in mathematical research projects
- Served on curriculum development committee for mathematics program

ACADEMIC SCHOLARSHIP/RESEARCH/CREATIVE ENDEAV- ORS

Research Interests

- Agent-Based Modeling: Crowd Dynamics and Crime Pattern Analysis
- Pattern Formation and Parameter Learning in Complex Systems
- Numerical Optimization and Scientific Computing
- Mathematical Modeling of Social Phenomena

Publications

1. Daewa Kim, Demetrio Labate, **Kamrun N. Mily**, Annalisa Quaini. (2025). “Data-driven learning to enhance a kinetic model of distressed crowd dynamics.” *Mathematical Models and Methods in Applied Sciences*, 35(7), 1609–1636.
DOI: [10.1142/S021820252540007X](https://doi.org/10.1142/S021820252540007X)
2. Baoli Hao, **Kamrun N. Mily**, Annalisa Quaini, Ming Zhong. (2026). “A finite element framework for simulating residential burglary in realistic urban geometries” *Mathematical Models and Methods in Applied Sciences*, <https://doi.org/10.48550/arXiv.2508.11055>
(Accepted)

Conference Presentations and Activities

- **Attendee and Presenter**
SIAM TX-Louisiana Sectional Conference, UT Austin, September 26–28, 2025
- **Attendee and Presenter**
Graduate Student Paper Presentation (GSPP 2026), a one-day, conference-style event organized by the AMS and SIAM Student Chapters to talk about recent developments in research in Mathematics.
- **Poster Presentation**, “Data-driven Learning to Enhance a Kinetic Model of Distressed Crowd Dynamics” in CBMS Conference, Department of Mathematics, University of Houston, December 8–12, 2025

- **Co-organizer**, Minisymposium on Agent-Based Modeling
SIAM TX-Louisiana Sectional Conference, University of Texas at Austin, September 26–28, 2025
- **First Place Winner**, Three-Minute Thesis (3MT) Competition
AMS Student Chapter, University of Houston, 2025

PROFESSIONAL DEVELOPMENT AND TRAINING

- **ChAMELEON Summer School 2025**
University of Houston – Advanced training in computational mathematics
- **TYRIT (Transfer Your Research Into Teaching) Course Design Workshop**
Graduate School, University of Houston – Pedagogical training for research integration
- **Professional Development Workshop Series**
Life Carnival Foundation – Goal Setting, Presentation Skills, and Professional Growth

LEADERSHIP AND SERVICE

University Service

- **Treasurer**, Association for Women in Mathematics (AWM) Chapter
University of Houston, Spring 2024 – Spring 2025
– Managed financial records and transactions for chapter activities
– Coordinated fundraising initiatives for student professional development
- **Communication Chair**, AWM Chapter
University of Houston, Fall 2022 – Fall 2023
– Developed communication strategies and maintained chapter website
– Organized networking events and professional development workshops

Professional Service

- Conference Minisymposium Co-organizer, SIAM TX-LA Section, 2025
- Peer Reviewer for graduate student research symposium, 2024–Present
- Volunteer Tutor, Houston-area mathematics outreach programs, 2022–Present

TECHNICAL SKILLS

- **Programming Languages:** Python (Advanced), MATLAB (Advanced), R (Intermediate)
- **Mathematical Software:** Mathematica, SageMath, COMSOL
- **Document Preparation:** LaTeX, Beamer, Jupyter Notebooks
- **Data Analysis:** NumPy, SciPy, Pandas, Matplotlib
- **Other:** Microsoft Office Suite, Canvas LMS, Git/GitHub

HONORS AND AWARDS

- Graduate Tuition Fellowship, University of Houston, 2021 – Present
- Teaching Assistantship Award, University of Houston, 2021 – Present
- Bangladesh Sweden Trust Fund Fellowship, Ministry of Finance, Bangladesh, 2021
- A.F. Mujibur Rahman Gold Medal (First Class First, MS), University of Dhaka, 2017